Kapasa Makasa University

Cybersecurity, Year III, 2024

System Vulnerability Assessment

Cys 330

End of Academic year leaving Examinations

Date: 14/10/2024, 09-12 HOURS **Duration: 3 Hours**

Instructions to candidates

1. Write your **Name** and **Student ID** on the **Answer Sheet** provided
2. There are **Seven (7) questions** in this paper. Attempt only **five (5). Question One** is compulsory and choose **any four (4)** from the remaining **seven (7).** Indicate on the Answer sheet your choice from the optional questions.
3. Write your Answers on the answer Sheet provided.

Information to candidates

No Foreign material will be allowed in this examination.

The number of marks is shown in brackets **[ ]** at the end of each question or part question.

Clear handwriting and neat work is encouraged.

The total number of marks for this paper is **100**

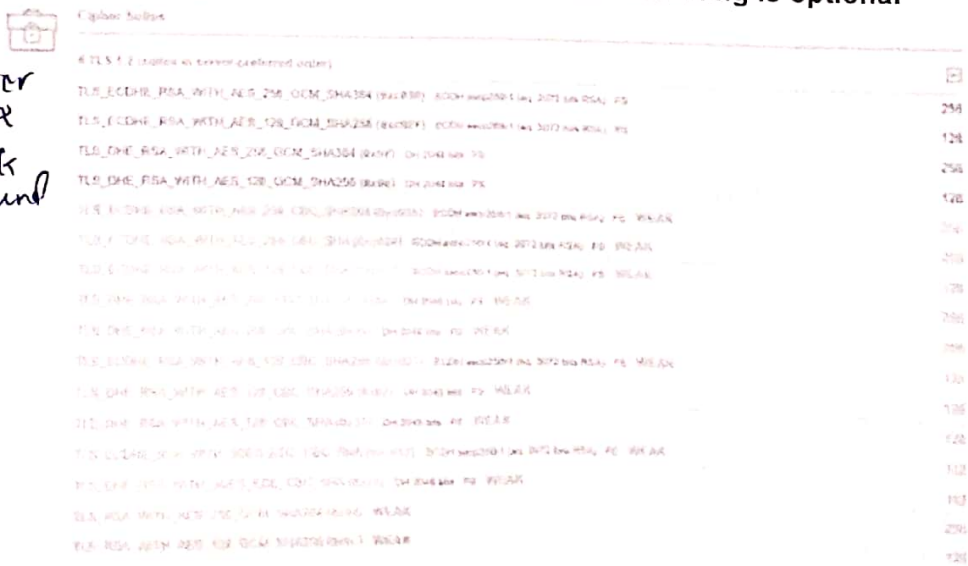
DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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Question One [20 Marks] Compulsory

1. Define the following vulnerability assessment terms:
   (a) Typosquatting [1]
   (b) Zero-day vulnerabilities [1]
   (c) Window of vulnerability [1]
   (d) Cybersquatting [1]
2. Mention **three (3)** characteristics of a zero-day vulnerability [3]
3. As an IT Cybersecurity, you have scanned the system for the organisation you work for. The screenshot below shows the results of the scan.
   (a) Write a Technical Report based on the real time scan on the organisation system. Below, is the guide that could be useful for your answer.
   (i) Title [1]
   (ii) Introduction [1]
   (iii) Methodology [2]
   (iv) Findings [3]
   (v) Hypothesis [2]
   (vi) Recommendations [3]
   (vii) Conclusion [1]

Note that using this Screenshot for a real time scanning is optional



Optional Questions
Instructions

1. **Choose any other four (4) questions from the six (6) questions below. Note that all questions have the same mark allocations [20 Marks].**
2. **Indicate the choice of your answer on the Answer Booklet provided.**

Question Two [20 Marks]

1. Compare passive and active vulnerability analysis in terms of the following;
   (a) Scalability [1]
   (b) Accuracy [1]

perform

(c) Effectiveness [1]
(d) Deployment [1]
(e) Resource use [1]
(f) Security vulnerabilities [1]

2. (a) List and explain **two (2)** advantages of Active vulnerability analysis [2]
   (b) List and explain **two (2)** disadvantages of Active vulnerability analysis [2]
   (c) Give and explain **two (2)** advantages of Passive vulnerability analysis [2]
   (d) List and explain **two (2)** disadvantages of Passive vulnerability analysis [2]
3. Compare Internal and External scanners in terms of the following:
   (a) Objectives [1]
   (b) Proactive and reactive [2]
   (c) Scope [1]
   (d) Use cases [1]
   (e) Location [1]

Question Three [20 Marks]

1. Demonstrate the environment in which the following vulnerability assessment technologies below are used;
   (a) Monolithic architecture [2]
   (b) Agent based architecture [2]
   (c) Cloud based architecture [2]
   (d) Multi – source integration [2]
   (e) Agentless based architecture [2]
2. Differentiate between Agent and Agentless architecture [2]
3. Explain possible challenges you are likely to face with an Agent based architecture in terms of false positives, management complexity, scalability concerns and thread vector. [4]
4. Write short notes on monolithic architecture with focus on complexity and rigidity, updating, attack surface and flexibility. [4]

Question Four [20 Marks]

1. Explain briefly in short note form, on hardware vulnerability [2]
2. Write short notes on the following focusing on the explaining what it is and also, the impact and mitigation
   (a) susceptibility to humidity and dusk [3]
   (b) Natural disasters [3]
   (c) Soiling and contamination [3]
   (d) Firmware [3]
3. Discuss software vulnerabilities in terms of;
   (a) Design flaws, impact and mitigation [3]
   (b) Memory safety violations, impact, and mitigation [3]

Question Five [20 Marks]

1. Assess the following with regard to Common Vulnerability Scoring System (CVSS) and also, indicating their colour codes at the end using short notes
   (a) Critical 9.0 - 10.0 red [3]
   (b) High 7.0 8.9 orange [3]
   (c) Medium 4.0 6.9 yellow [3]
   (d) Low 0.0 3.9 light green [3]
2. Evaluate the following metric groups below;

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(a) Base metrics [1]
  (i) Exploitability metrics [½]
  (ii) Attack vector [½]
  (iii) Attack complexity (AC) [½]
  (iv) User Interaction (UI) [½]
(b) Temporal metrics [1]
  (i) Exploit code maturity [½]
  (ii) Remediation level [½]
(c) Environmental metrics [1]
  (i) Modified base metrics [½]
  (ii) Security requirements [½]

3. Defend the need for Air-gapping in dealing with malware vulnerabilities in an organisation [1]

Question Six [20 Marks]

1. With the aid of a diagram, formulate and explain a Vulnerability Management Lifecycle [[12]
2. Outline **six (6)** bests practices for the act stage in VML [6]
3. Explain the importance of the Tenable Security Centre [2]

Question Seven [20 Marks]

1. Attached, is data to help you answer the questions that follow.

COMPANY DETAILS

- **Company name:** Kapasa Makasa University
- **Location:** Chinsali, Muchinga Province
- **Industry:** Science and Technology
- **Company Size:** 950 Employees

Virtual Statics
VCDFR

TEST DETAILS

- **Test Start Date:** 1st October 2024
- **Test Duration:** 7 days
- **Network Analysed:** Internal Local Area Network (LAN)
- **Functions enabled:** IPS/AV/Web/APP ctrl

CHALLENGES

- **IPS Attacks detected:** 1,1002
- **Malware and Botnet events detected:** 32
- **High Risks applications detected:** 200
- **Application categories:** Network Service /video/Audio/Web/Others
- **Top Web categories:** Facebook and Instagram/ streaming media and downloading / web-based emails.
- **Top 3 web domains:** Mail. Google.com / stream pandora.com / en.wikipedia.org

NETWORK UTILIZATION

- **Top Hosts/Clients by Bandwidth** 192.0.2.0.24/ 102.130.100.0/102.130.100.255 / 102.144.0.0
- **Average throughput:** 28 Mbps
- **Unique Hosts detected:** 503

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ADDITIONAL INFORMATION

- Evasive applications: 23
- P2P and filesharing applications: 10
- Bandwidth consuming applications: 32

Based on the information given above, as a Cybersecurity Expert, write a comprehensive Technical Report for Kapasa Makasa University

Helpful information in no particular order

- Scan results
- Executive Summary [3]
- Risk Assessments [3]
- Recommendations [3]
- Findings [4]
- Methodology [3]

2. Highlight the differences between Macro level and Micro – Level [2]

broader small [2]

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